Intelligent Staff Email Personalization & Automation System

Project progress report

Repository	github.com/LibiniP/intelligent-staff-email-automation (branch main)
Report date	24 October 2026
Status	Working prototype. Runs end to end on mock data. Outlook sending and real login not yet connected.
Tests	25 automated tests passing

1. Summary

The project builds an internal platform that finds staff occasions such as birthdays, joining anniversaries, service milestones, Silver Jubilee and retirement. It then drafts personalized emails from verified institutional data, sends them through a human approval step, schedules them and delivers them through Microsoft 365 / Outlook, keeping a full audit trail.

The core pipeline is built and works end to end on mock data supplied for testing: 320 staff were imported and a scan produced correct drafts for every occasion type, with the approval, scheduling and audit steps all working. Two parts of the brief remain open and are described in section 6: live Outlook sending and secure staff login.

Pipeline:

Staff data	Milestone detection	Rules	Context	Draft	Approval	Schedule	Send	Audit

2. What was done

Step	Work
1. Blueprint	Read the project brief and produced the architecture, tech stack, module list, data model, a 12-week phase plan and a list of decisions needed outside the code.
2. First build	Backend API, database models, milestone engine, rules, templates, guarded drafting, approval flow, scheduler, Outlook sender, audit log, analytics, and an admin dashboard.
3. Data and team setup	CSV and Excel import, Postgres via Docker, configuration slots for every API key in `.env.example`, and setup instructions for a teammate.
4. Handover files	`CLAUDE.md` (project rules and code map) and a code graph in `graphify-out/` so a new developer or AI assistant can understand the code quickly.
5. Debugging	Three rounds of review. Ten bugs found and fixed, each with a regression test.
6. Real data layout	Importer adapted to the supplied Excel file (three sheets, 320 staff). A `retirement_date` column was added to the mock file and the code updated to use it.
7. Screenshots	Four dashboard screenshots added to the repository README.

Five commits are on the main branch, four of them from this work. Real staff data, the .env file and databases are excluded from git by design.

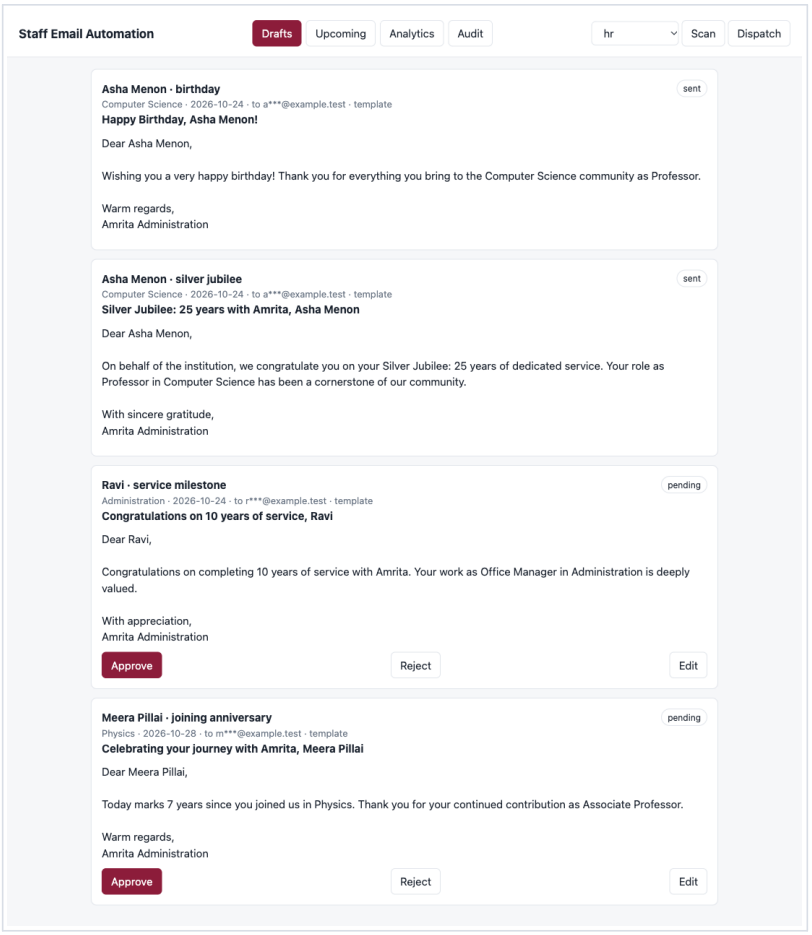
3. What the software does

Stage	Behaviour
Staff data	Imports Excel or CSV. Reads every sheet that has a staff ID column, recognises common header names, and skips bad rows with a reason. Re-importing updates people and never duplicates them. A failed import rolls back completely.
Who gets email	Active and on-leave staff only. Retired and resigned staff are kept but skipped. Opted-out staff are skipped, checked at scan time and again at send time.
Detection	Birthdays, joining anniversaries, service milestones (5, 10, 15, 20, 30, 35, 40 years), Silver Jubilee (25 years) and retirement. Handles 29 February and New Year boundaries. After a person's retirement date they receive no birthday or anniversary mail.
Rules	Per occasion: enabled or not, days ahead, approval required, send hour. Stored in the database.
Drafting	Fills a template with verified facts only: name with title, department, designation, years of service. Unknown placeholders are rejected. An optional AI polish step is discarded if it adds any number or drops the name, and the template text is used instead.
Approval	Reviewers approve, reject, edit or cancel. Retirement, promotion and Silver Jubilee always need a human, whatever the rule says. Birthdays can auto-schedule.
Scheduling and send	Approved drafts send at the set hour (India time) through Microsoft Graph. Status is saved after each email so a crash cannot cause a resend. Failures are recorded.
Safety	One email per person per occasion per year, enforced by a database rule. Recipient addresses are masked in the dashboard. Every action is written to the audit log.
Dashboard	Drafts, Upcoming, Analytics and Audit tabs, with Scan and Dispatch buttons.

4. Results on the mock data

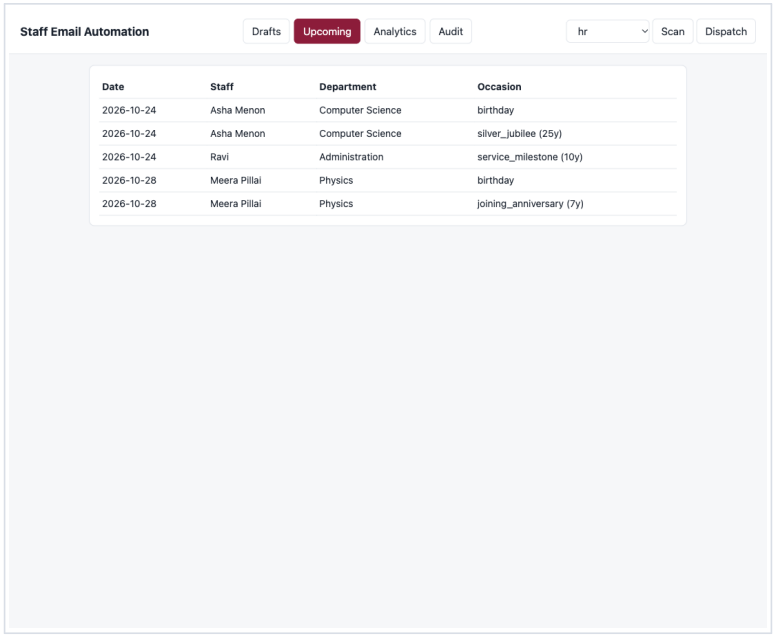
- 320 staff imported (189 faculty, 69 non-teaching, 62 admin) with 0 skipped rows. 299 are active.
- A scan on 24 October 2026 produced 15 drafts: 5 birthdays (auto-scheduled), 4 joining anniversaries, 4 service milestones, 1 Silver Jubilee and 1 retirement (all pending approval).
- Generated text was correct, for example: "Dear Ms. Radhika Chatterjee ... 25 years of dedicated service".
- Approving a retirement draft moved it to scheduled, and the audit log recorded who did what and when.
- 25 automated tests pass with warnings treated as errors, and the static checker reports no problems.

Drafts screen: review, edit, approve or reject (demo data only).

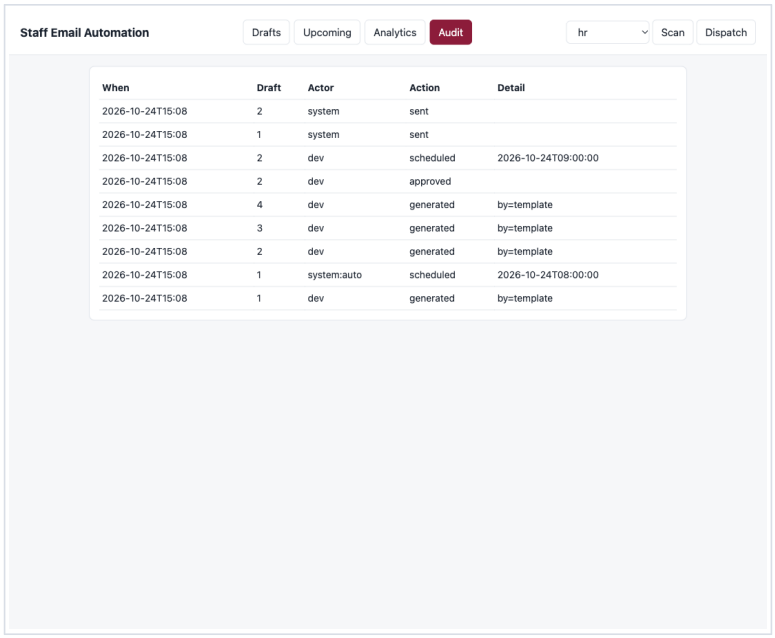


Other screens (demo data only)

Upcoming occasions



Audit log



5. Quality: bugs found and fixed

Each fix has an automated test. The two concurrency tests were confirmed to fail on the old code.

Bug	What would have gone wrong
Template typo sent as-is	A saved template containing a mistyped placeholder would email staff literal text such as "Dear $nmae". Now rejected on save and at draft time.
One collision undoes a scan	A duplicate from a parallel scan rolled back every draft made earlier in that scan. Each draft is now saved independently.
Bad rule values crashed server	Send hour 25 or negative days caused server errors. Inputs now validated.
Resend after a crash	Sent status was saved once at the end, so a crash could resend delivered mail. Now saved after each email.
No headers meant admin	A request with no role header received full admin rights. Now rejected.
Timestamps mismatched	Audit log used UTC, send times used India time. All now use one configured timezone.
Opt-out reset on re-import	Re-importing a file without an opt-out column cleared existing opt-outs.
Missing rule crashed approval	Approving a draft whose rule row was deleted crashed. Now falls back to 9 am.
Half-imported files	A bad sheet partway through an Excel file left earlier sheets half-imported. Now rolls back everything.
Old databases crashed	A database from before the new staff fields would fail. Missing columns are now added automatically at startup.

6. Mapping to the brief and open items

Brief module	Status
Staff directory, milestone engine, rules engine, templates	Done
Personalization and AI-assisted generation, hallucination guard	Done

Brief module	Status
Approval dashboard, scheduling, communication history	Done
Analytics dashboard	Basic counts only (by status and by occasion)
Access control and privacy	Partly. Role checks exist, but login is a development placeholder.
Microsoft 365 / Outlook integration	Code written, not tested against a real tenant

Open items and known limits

- **Login.** Roles come from request headers, so anyone could claim admin. Entra ID login must be built before real use.
- **Outlook.** Sending runs in mock mode. It needs the IT-approved Azure app registration (only the Mail.Send permission, restricted to one sender mailbox) and a test mailbox.
- **Database.** Tested on SQLite only. Postgres is configured but has not been run. No formal migrations yet.
- **Data.** The Excel file is mock data. Its retirement dates follow a made-up rule (last day of the month of the 60th birthday). Real dates must come from HR and must never be computed in code.
- **Automation.** No background worker yet. Scan and dispatch run from buttons or the command line.
- **Missing features.** Promotion and institutional-event emails are not built. The dashboard has no template or rule editing screens, although the API for them exists.
- **Late sends.** A draft approved after its date sends immediately, so a birthday email could arrive late. Whether that is acceptable is a policy decision for HR.
- **Failed sends** are logged but not retried automatically.
- **Two dispatchers at once** could double-send. Safe with one worker; needs row locking once a background worker exists.

7. Recommended next steps

#	Step	Why
1	Entra ID login	Required for security before any real use.
2	Azure app registration and test-mailbox send	Proves the Outlook integration works.
3	Run on Postgres with the real data source	Confirms the production database and real columns.
4	Background worker	Makes scan and send automatic and safe.
5	Template and rule editing screens	Lets HR manage wording and timing without a developer.
6	Confirm policy questions with HR	Late-send behaviour, retirement dates, opt-out handling, LLM data policy.